Reader: The converse theorem is not true. For example, sequence (10) which was discussed in the first dialogue is bounded. However, it has no limit.
(Sera answer)

Author: Right you are. We thus come to a

Corollary: The boundedness of a sequence is a necessary condition for its convergence; however, it is not a sufficient condition. If a sequence is convergent, it is bounded. If a sequence is unbounded, it is definitely nonconvergent.

Imp.

Seq 10: $1, \frac{2}{3}, \frac{1}{3}, \frac{4}{5}, \frac{1}{5}, \frac{6}{7}, \frac{1}{7}, \frac{8}{9}, \ldots\ldots$

Reader: I wonder whether there is a sufficient condition for the convergence of a sequence?

Author: We have already mentioned this condition in the previous dialogue, namely, simultaneous validity of both the boundedness and monotonicity of a sequence.

The Weierstrass theorem states: If a sequence is both bounded and monotonic, it has a limit.

- Unfortunately, the proof of the theorem is beyond the scope of the book; we shall not give it. I shall simply ask you to look again at sequences (5), (7) and (13) (see Dialogue One), which satisfy the conditions of the Weierstrass theorem.

Seq 5: $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \frac{6}{7}, \frac{7}{8}, \frac{8}{9}, \ldots\ldots$

Seq 7: $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \frac{1}{7}, \frac{1}{8}, \ldots\ldots$

Seq 13: $3, 3.1, 3.14, 3.141, 3.1415, 3.14159, \ldots\ldots$

Reader: As far as I understand, again the converse theorem is not true. Indeed, sequence (9) (from Dialogue One) has a limit but is not monotonic.

Seq 9: $1, -1, \frac{1}{3}, -\frac{1}{3}, \frac{1}{5}, -\frac{1}{5}, \frac{1}{7}, -\frac{1}{7}, \ldots\ldots$

Author: That is correct. We thus come to the following

Conclusion: If a sequence is both monotonic and bounded, it is a sufficient (but not necessary) condition for its convergence.

Reader: Well, one can easily get confused.